

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4341	Course Title: Mathematical Foundations for ML
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide a mathematical introduction in the field of Linear Algebra, calculus, probability and statistics to understand and apply algorithms in various applications.

Course Prerequisites:

Course name	Semester
1002- Mathematics I	1
2002- Mathematics II	2

Course Outcomes

On completion of the course, the students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Solve the system of linear equations and implement Eigenvalues and Eigenvectors in engineering problems	13	Applying
CO2	Utilize the concept related to partial derivatives to solve problems of maxima and minima of multivariable functions.	15	Applying
CO3	Apply the concepts of probability for solving problems.	15	Applying
CO4	Assess data using statistical methods and analyze that data by applying hypothesis testing methods.	15	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Solve the system of linear equations and implement Eigenvalues and Eigenvectors in engineering problems		
M1.01	Explain the concept of vector spaces	2	Understanding
M1.02	Describe linearly independent and dependent vectors	2	Understanding
M1.03	Compute the rank of a matrix	4	Applying
M1.04	Solve a system of linear equations by Gauss elimination method and evaluate the EigenValues and EigenVectors of a matrix	5	Applying
Contents: Vector spaces: Vector spaces, linear independence, basis and rank, linear combination of vectors, linearly independent and dependent vectors, rank of a matrix Systems of linear equations:- Method of solution:- Gauss Elimination Method ,Eigenvalues and Eigenvectors and their properties.			
CO2	Utilize the concept related to partial derivatives to solve problems of maxima and minima of multivariable functions.		
M2.01	Illustrate the Concept of limit and continuity of functions of two variables	4	Applying
M2.02	Outline the Concept of first and second order Partial derivatives	4	Applying
M2.03	Explain the Concept of maxima and minima of functions of two variables and apply them in related problems	4	Applying

M2.04	Demonstrate Multivariate differentiation	3	Applying
	Series Test – I	1	
Contents: Calculus: Introduction to Functions, Limits and Continuity, Differentiability, Derivative rules and operations, partial derivatives, maxima and minima of functions of two variables, Multivariate differentiation (Simple problems based on these topics).			
CO3	Apply the concepts of probability for solving problems.		
M3.01	Demonstrate the basic concepts in sample space, dependent and independent events	2	Applying
M3.02	Illustrate conditional probability and Bayes' Theorem	3	Applying
M3.03	Compare the difference between Discrete and continuous random variables	3	Applying
M3.04	Calculate the expected value and variance of discrete and continuous random variables	3	Applying
M3.05	Identify the characteristics of discrete probability distributions- binomial and poisson	2	Understanding
M3.06	Use probability distributions to solve simple problems	2	Applying
Contents: Probability: Axiomatic definition - sample space, dependent and independent events conditional probability, Bayes' Theorem Random variables: Discrete and continuous random variables, expectation, variance, Probability distributions- binomial, poisson			
CO4	Assess data using statistical methods and analyze that data by applying hypothesis testing methods.		
M4.01	Describe the fundamental operations to be performed on data	1	Understanding
M4.02	Calculate the measures of central tendency and dispersion	3	Applying
M4.03	Explain the concept of Sampling and Sampling Distributions	3	Understanding
M4.04	Describe the types of errors	2	Understanding
M4.05	Calculate confidence interval for sample mean	3	Applying
M4.06	Demonstrate the concept of hypothesis testing	3	Applying

	Series Test – II	1	
Contents: Fundamentals of Data: Collection, Summarization, and Visualization, Measures of central tendency and dispersion, Sampling and Sampling Distributions, Types of Errors, level of significance, Confidence Interval Estimation and Hypothesis Testing for Z -test and t- Test.			

Text / Reference:

T/R	Book Title / Author
T1	H. Anton, I. Biven, S. Davis, "Calculus", Wiley, 10th edition, 2015.
T2	Mathematics for Machine Learning by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong published by Cambridge university press
T3	Linear Algebra and its applications, 4th edition by Gilbert Strang
T4	Introduction to Applied Linear Algebra by Stephen Boyd and Lieven Vandenberghe, 2018 published by Cambridge university Press
T5	Jay L. Devore, probability and Statistics form Engineering and the Science S, 8th edition, Cengage, 2012

Online Resources:

Sl. No.	Website Link
1	https://www.w3schools.com/ai/ai_mathematics.asp
2	https://mml-book.github.io/book/mml-book.pdf